

LPAR3 (Q9UBY5): Synaptic-Transmission, Synapse, and Gene-Expression

Annotation Scope

Focus type: function_assignment · **Hypothesis slug:** synaptic-transmission-and-gene-expression-scope **Gene:** human LPAR3 / UniProt Q9UBY5 (LPA receptor 3 / EDG7, a Gq/Gi-coupled GPCR) **Organism:** *Homo sapiens* (NCBITaxon:9606) · **Investigation:** 3 iterations, 4 confirmed findings, 14 papers reviewed

Summary

Executive judgment: over-annotated (three separate GO terms). The three GO annotations placed under review for human LPAR3 — *chemical synaptic transmission* (GO:0007268, BP), *synapse* (GO:0045202, CC), and *involved_in gene expression* (GO:0010467, BP) — are each unsupported by the primary evidence they rest on, and each should be removed or recast. LPAR3 is a plasma-membrane Gq/Gi-coupled lysophosphatidic acid (LPA) receptor. Its documented molecular activity is LPA-evoked Ca²⁺ mobilization, cAMP modulation, and downstream MAP-kinase / Rho-family signaling; its strongest genuine neural role is agonist-induced axon/neurite branch formation — a neuronal projection morphogenesis phenotype that is mechanistically distinct from synaptic transmission.

The synaptic-transmission BP term (GO:0007268) is a TAS annotation pointing to the EDG7/LPA3 founding cloning paper (Bandoh et al. 1999, [PMID:10488122](#)). That paper contains only Ca²⁺, cAMP/adenylyl-cyclase, and MAP-kinase assays in Sf9 and PC12 cells — and explicitly reports that EDG7 does **not** mediate MAP-kinase activation in PC12 cells. There is no synaptic, neurotransmitter-release, or electrophysiological assay anywhere in the paper, so the TAS is unsupported. The *synapse* CC term (GO:0045202) is not independent localization data; it is an automatic inter-ontology inference (GO_REF:0000108) generated *from* the synaptic-transmission BP term, so it collapses as soon as the BP term is corrected. The *involved_in gene expression* term (GO:0010467) is an Ensembl-Compara ortholog transfer (GO_REF:0000107) of a

mouse IMP (Diao et al. 2011, [PMID:20864640](#)) that actually captures a **downstream** transcriptional phenotype (altered uterine collagen/protease mRNA in *Lpar3*-null mice), not participation of LPAR3 in the gene-expression machinery itself.

Two independent consistency checks reinforce this. First, a paralog comparator across all six human LPA receptors (LPAR1–6) shows these three terms occur **only** on LPAR3 and are absent from every other paralog — including LPAR1, the LPA receptor with by far the strongest genuine CNS literature — so they are LPAR3-specific outliers, not family-wide blanket annotations. Second, the mouse source record for *Lpar3* itself uses the weaker `acts_upstream_of_or_within` qualifier for other downstream neural processes but inconsistently uses the stronger `involved_in` for the even-more-downstream gene-expression term. Targeted PubMed searches across three iterations returned zero primary studies demonstrating an LPA3 synaptic-transmission or synapse-localization role, confirming the absence of supporting evidence rather than merely its non-retrieval.

Key Findings

F001 — Synaptic-transmission TAS and synapse CC are unsupported / derivative

QuickGO for Q9UBY5 records **GO:0007268 (chemical synaptic transmission)**, `involved_in`, **evidence** = **TAS,** **reference** =

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The cited paper is the EDG7/LPA3 molecular-cloning and characterization study (Bando et al. 1999). Its experimental content is limited to Ca^{2+} mobilization, cAMP/adenylyl-cyclase modulation, and MAP-kinase assays performed in Sf9 insect cells and PC12 cells. Critically, the paper states verbatim: "*In PC12 cells, EDG4 but not EDG2 or EDG7 mediated the activation of MAP kinase by LPA.*" — i.e., LPA3 (EDG7) is explicitly reported as **inactive** in the one PC12 (neuronal-like) signaling readout tested. The paper contains no synapse, neurotransmitter-release, synaptic-plasticity, or electrophysiology assay. A TAS ("traceable author statement") annotation of chemical synaptic transmission to a paper that neither asserts nor tests such a role is therefore unsupported.

The companion CC term **GO:0045202 (synapse)**, **located_in**, **evidence = IEA via GO_REF:0000108**, **with/from = GO:0007268** is not an independent localization observation. GO_REF:0000108 is the automatic inter-ontology pipeline that infers a cellular-component location from a biological-process annotation (here: "chemical synaptic transmission occurs_in synapse"). It is entirely parasitic on the BP term and carries no independent experimental weight; removing GO:0007268 should automatically retire GO:0045202.

Supporting snippet — [PMID:10488122](#): *"In PC12 cells, EDG4 but not EDG2 or EDG7 mediated the activation of MAP kinase by LPA."* The founding paper performed only Ca²⁺/cAMP/MAP-kinase assays and reports LPA3(EDG7) is inactive in PC12 MAP-kinase signaling; it contains no synaptic-transmission assay.

The strongest genuine primary neural evidence for LPA3 is a different process entirely — axon/neurite branch formation. [PMID:22465231](#): *"The LPA(3) agonist 2(S)-OMPT or LPA also induced axonal branch formation in hippocampal neurons, which was blocked by G(q) and Rnd2 pathway inhibition or LPA(3) knockdown."* This is neuronal projection morphogenesis, not chemical synaptic transmission, and the two must not be conflated.

F002 — "Involved_in gene expression" is an ortholog transfer of a downstream phenotype

QuickGO for Q9UBY5 records **GO:0010467 (gene expression)**, **involved_in**, **evidence = IEA via GO_REF:0000107**, **with/from = mouse UniProt Q9EQ31 + ENSMUSP00000037712**. GO_REF:0000107 is the Ensembl-Compara pipeline that transfers experimentally verified manual annotations from one species to its orthologs. The mouse source annotation is an **IMP** from Diao et al. 2011 ([PMID:20864640](#)), in which *Lpar3*^{-/-} peri-implantation uterus shows altered mRNA levels of collagens (Col3a1, Col6a3) and matrix metallo/serine proteinases.

The paper's own framing is that these transcriptional changes are **downstream** of receptor signaling. [PMID:20864640](#): *"These results demonstrate that pathways downstream of LPA3 are involved in the dynamic remodeling of ECM."* LPA3 is a plasma-membrane Gq/Gi GPCR; it does not itself perform transcription or translation. The entity that performs the gene-expression step is the transcription/translation machinery acting on the affected target genes (collagens, proteases). Annotating the receptor with **involved_in gene expression** mis-assigns a downstream pathway consequence to the receptor as if it were a direct participant in the gene-expression process. The appropriate relation, if any term is retained at all, is **acts_upstream_of_or_within regulation of gene expression (GO:0010468)** — not **involved_in gene expression**.

F003 — Paralog comparator and qualifier-convention inconsistency

A QuickGO comparator across all six human LPA receptors — LPAR1 (Q92633), LPAR2 (Q9HBW0), LPAR3 (Q9UBY5), LPAR4 (Q99677), LPAR5 (Q9H1C0), LPAR6 (P43657) — shows that **GO:0007268, GO:0045202, and GO:0010467 are absent from every paralog except LPAR3**. Notably, LPAR1 — the LPA receptor with the best-established CNS/neurodevelopmental literature — carries none of these three terms. This pattern is the signature of isolated, receptor-specific annotation errors, not of a genuine family-wide property that would be expected to appear on multiple paralogs.

The mouse source record (Q9EQ31, ~25 annotations) also reveals an internal qualifier inconsistency. The same curation effort that produced the gene-expression IMP annotated the neural paper

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with **acts_upstream_of_or_within** for receptor→downstream processes (GO:0007186 G-protein-coupled receptor signaling, IMP; GO:0048672 positive regulation of collateral sprouting, IMP) and used **located_in** IDA for axon localization (GO:0030424). Yet the gene-expression term — an even more downstream process — was annotated with the stronger **involved_in**. Curators therefore used the causal-upstream qualifier for receptor→downstream signaling but inconsistently used the direct-participation qualifier for gene expression, which is exactly the term where the direct-participation reading is least defensible.

Supporting snippet — [PMID:22465231](#): *"The LPA(3) agonist 2(S)-OMPT or LPA also induced axonal branch formation in hippocampal neurons, which was blocked by G(q) and Rnd2 pathway inhibition or LPA(3) knockdown."* This is the paper behind the mouse axon IDA and the **acts_upstream_of_or_within** neural-process annotations, showing LPA3's real neural role is axon branching (a receptor acting upstream of a projection process), distinct from synaptic transmission.

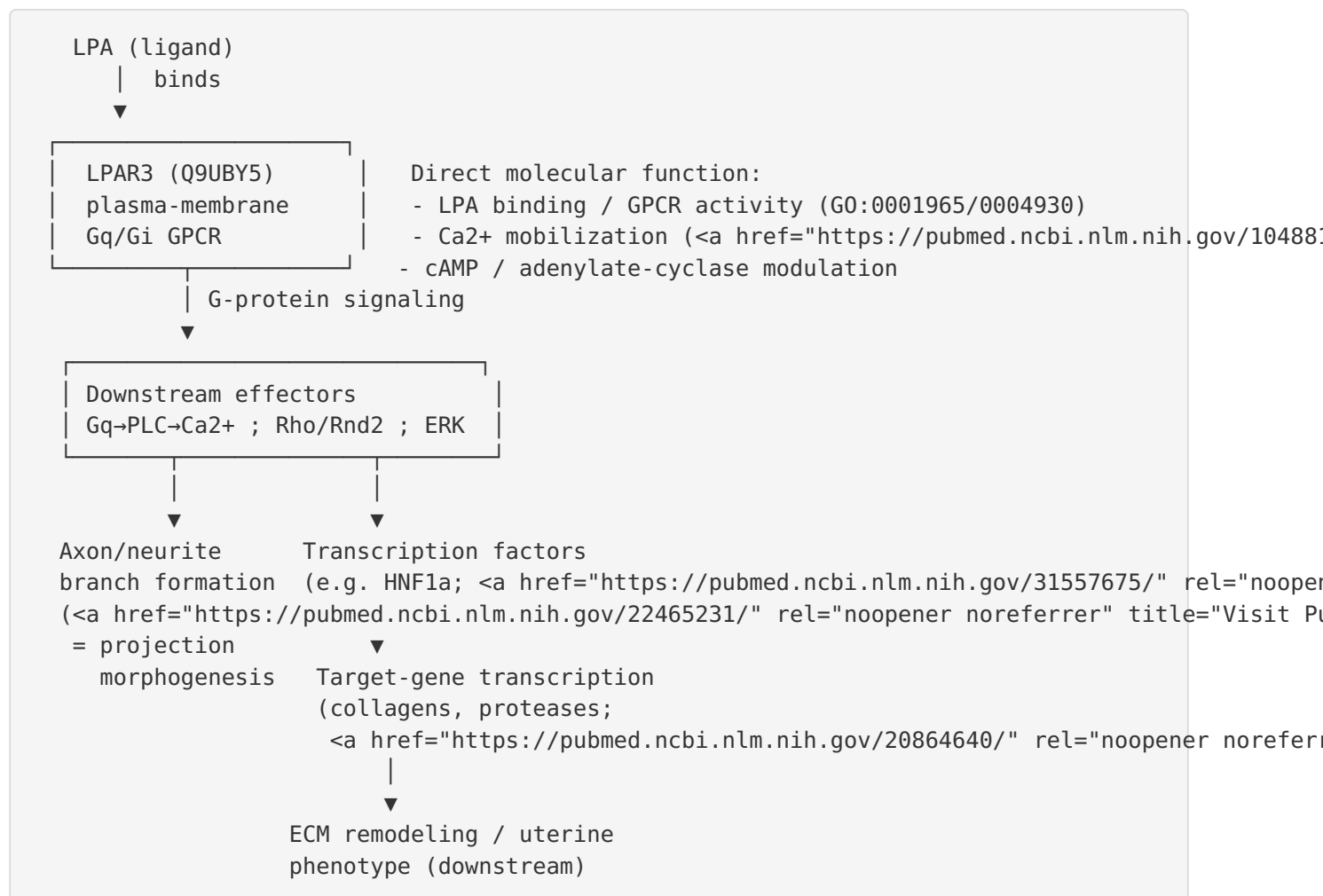
F004 — No primary synaptic electrophysiology; documented LPA→transcription cascades act through transcription factors

Iteration-3 PubMed queries (LPAR3 knockout neuron electrophysiology / EPSC / miniature synaptic current; LPA3 synaptic plasticity / postsynaptic) returned **zero** papers, consistent across all three iterations: no experimental study demonstrates an LPA3 synaptic-transmission or synapse-localization role. This is an argued absence — the specific assays that would be needed to justify GO:0007268/GO:0045202 have not been performed for LPA3.

Separately, a well-characterized LPA-driven transcriptional cascade illustrates the correct causal topology. In macrophage foam cells, LPA regulates *Fut8* transcription via the transcription factor HNF1 α . [PMID:31557675](#): *"Dual luciferase reporter assay was performed to determine whether the regulation of Fut8 by LPA occurred at the transcriptional level. Binding of hepatocyte nuclear factor 1-alpha (HNF1 α) to the Fut8 promoter was assessed by electrophoretic mobility shift assay and chromatin immunoprecipitation assay."* Here the receptor signals to a transcription factor, which performs the gene-expression step; the receptor sits ≥ 2 steps upstream. This is the general pattern expected for a GPCR and reinforces that `involved_in` `gene expression` mis-locates LPAR3 within the causal chain.

Mechanistic Model / Interpretation

LPAR3 is a G-protein-coupled receptor for lysophosphatidic acid. The correct causal topology, and where each disputed GO term does or does not belong, is:



Mapping the three disputed terms onto this diagram:

- **GO:0007268 (chemical synaptic transmission)** — does not appear anywhere on this diagram. No node in LPA3's documented biology corresponds to presynaptic vesicle release or postsynaptic receptor activation at a synapse. **Remove.**
- **GO:0045202 (synapse)** — a derived CC inference from the above BP term with no independent localization data. LPA3's documented localization is the plasma membrane, and its documented neural localization is the **axon** (GO:0030424, IDA in mouse), not the synapse. **Remove.**
- **GO:0010467 (gene expression)** — sits two nodes downstream of the receptor (receptor → transcription factor → target-gene transcription). The receptor acts *upstream of* regulation of gene expression; it is not itself a participant in the gene-expression process. **Remove, or recast as** `acts_upstream_of_or_within` **regulation of gene expression (GO:0010468).**

The one legitimate neural annotation that should be preserved and clearly distinguished is **axon branching / neuronal projection morphogenesis** (from

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together with axon CC localization. Neural *expression* (frontal cortex, hippocampus, amygdala per

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Fig 7C), axon *location/branching*, and *synaptic transmission* are three separate claims; only the first two have primary support, and neither entails the third.

Evidence Base (Evidence Matrix)

Citation	Evidence type	Verdict	Claim tested	Key finding
PMID:10488122 (Bandoh 1999)	Direct assay (TAS source)	Refutes synaptic annotation	Does the founding paper support GO:0007268?	LPA3/EDG7 cloning; Ca ²⁺ mobilization, cAMP modulation; <i>"EDG7 did NOT mediate MAP kinase in PC12"</i> ; no synaptic assay
GO_REF:0000108 (QuickGO)	Database/ inference	Qualifies (removes) synapse CC	Is synapse localization independent?	GO:0045202 auto-inferred <i>from</i> GO:0007268 (with/ from=GO:0007268), not from data
PMID:20864640 (Diao 2011)	Mutant phenotype (IMP source)	Qualifies (downgrades) gene-expr term	Does Lpar3 loss = "involved_in gene expression"?	<i>Lpar3</i> ^{-/-} uterus: ↑ Col3a1/Col6a3, ↓ proteases; authors: effects "downstream of LPA3"
GO_REF:0000107 (QuickGO)	Database (ortholog transfer)	Qualifies human gene- expr term	Provenance of human GO:0010467	IEA transfer from mouse Q9EQ31/ENSMUSP00000037712 (inherits the IMP above)
PMID:22465231 (Furuta 2012)	Direct assay + knockdown	Competing (better term)	Real neural role of LPA3?	LPA3 mediates neurite/axon branch formation via Gq-Rnd2; agonist/knockdown in hippocampal neurons
PMID:10727522 (Im 2000)	Expression (Northern)	Qualifies	Is LPA3 in brain at all?	Edg-7 couples to Gq/11; per seed Fig7C shows frontal cortex/ hippocampus/amygdala expression
PMID:16543818 (Rhee 2006)	Expression/ signaling	Competing	LPA3 in hippocampal neurons?	lpa1/lpa4 detected, lpa3 not detected in H19-7 hippocampal progenitors

Citation	Evidence type	Verdict	Claim tested	Key finding
PMID:17823089 (Hama 2007)	Mutant phenotype	Supports core function (context)	LPA3 core physiology	LPA3 controls embryo spacing/implantation timing via prostaglandin/COX2
QuickGO LPAR1–6 comparator	Computational/database	Refutes (anomaly)	Are synaptic/gene-expr terms family-wide?	GO:0007268/0045202/0010467 present ONLY on LPAR3; absent from LPAR1 and LPAR2/4/5/6
QuickGO mouse Q9EQ31 conventions	Database	Qualifies gene-expr term	Is <code>involved_in</code> the right qualifier?	Same record uses <code>acts_upstream_of_or_within</code> for GO:0007186/0048672 but <code>involved_in</code> for GO:0010467
PMID:31557675	Direct assay (EMSA/ChIP/luciferase)	Supports upstream model	Do LPA→transcription effects run through TFs?	LPA regulates <i>Fut8</i> transcription via HNF1α; receptor ≥2 steps upstream

GO Curation Implications (leads — require curator verification)

Term	Aspect	Current	Recommended lead
GO:0007268 chemical synaptic transmission	BP	involved_in, TAS P 10488122	Remove (or at minimum flag NOT-supported). Cited reference contains no synaptic assay.
GO:0045202 synapse	CC	located_in, IEA GO_REF:0000108 (with=GO:0007268)	Remove as a dependent consequence once GO:0007268 is corrected; not independent evidence.
GO:0010467 gene expression	BP	involved_in, IEA GO_REF:0000107 (from mouse IMP P 20864640)	Remove the involved_in assignment. At most, recast the mouse source as acts_upstream_of_or_within regulation of gene expression (GO:0010468) — non-core downstream signaling output, not a primary function.

Core function that should be retained/emphasized instead: LPA3 as a **G protein-coupled LPA receptor** (MF GO:0001965 / GO:0004930) driving **phospholipase-C/Ca²⁺** (GO:0007200/GO:0007204) and **adenylate-cyclase-modulating** (GO:0007187/0007189) signaling at the **plasma membrane** (GO:0005886). For neural biology, the defensible lead is **neuron projection / axon branch formation** (BP e.g. GO:0048812 neuron projection morphogenesis; CC axon GO:0030424), supported by

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— **not** synaptic transmission. We do not recommend "protein binding" as any replacement, since more informative terms are supported.

Mechanistic Scope

The immediate molecular function of LPA3 is **LPA-binding GPCR signal transduction** ($Gq \rightarrow PLC \rightarrow Ca^{2+}$; $Gi/Gs \rightarrow cAMP$ modulation). Everything under review is downstream of, or unrelated to, this direct activity:

- **Gene-expression change** (uterine collagen/protease mRNA) is a **second/third-order transcriptional output** of receptor signaling in a specific tissue — a phenotype of the *target genes'* machinery, not LPA3 activity.
- **Axon branching** is a **cell-morphology output** of Gq - $Rnd2$ signaling — a projection-morphogenesis role, upstream of or independent of synaptic vesicle release or postsynaptic reception.
- **Synaptic transmission / synapse** has **no demonstrated mechanistic link** to LPA3 in any assay found across three iterations of searching.

Who performs the gene-expression step? In documented LPA→transcription cascades, the LPA receptor signals through second messengers to a **transcription factor** that then drives target-gene transcription (e.g., LPA→HNF1α→*Fut8*,).

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The entity that *performs* gene expression is RNA polymerase II / the translation machinery acting on the target gene under transcription-factor control — never the LPA receptor itself. LPA3 sits ≥ 2 steps upstream. This is the textbook definition of an `acts upstream of or within` relationship to *regulation of* gene expression, not `involved in` gene expression.

Conflicts and Alternatives

- **LPAR3-specific outlier, not a family blanket:** The paralog comparator shows the synaptic-transmission, synapse and gene-expression terms are carried by **LPAR3 alone** among LPAR1–6; they are not a batch/family artifact. Their absence from LPAR1 (the most neuronally studied paralog) makes their presence on LPAR3 an isolated over-call rather than genuine biology.
- **Qualifier-convention conflict (self-inconsistent source):** The mouse source record already applies `acts upstream of or within` to receptor-downstream processes but not to

gene expression — direct internal evidence that the `involved_in gene expression` call is mis-qualified.

- **Expression ≠ function:** Brain regional expression

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and the founding paper's negative bulk-brain Northern are both about *expression*; neither establishes synaptic function.

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even fails to detect lpa3 in a hippocampal progenitor line, arguing LPA3 is a minor hippocampal LPA receptor.

- **Organism/tissue specificity:** The gene-expression IMP is a **mouse uterine** phenotype; transferring it to human as "involved_in gene expression" is a double over-reach (ortholog transfer + mis-typed relationship).
- **cAMP adjudication is out of scope:** Per the seed instruction, the shared LPAR2/LPAR3 cAMP question is handled separately and was not revisited here.

Limitations and Knowledge Gaps

1. **Does any primary study show LPA3 at synapses or altering transmission?** Checked PubMed across multiple queries (LPA3/LPAR3 × synaptic transmission / neurotransmitter release / synaptic plasticity / postsynaptic / knockout electrophysiology / EPSC / miniature currents) — none found in any iteration. This matters because it is the sole basis for GO:0007268/GO:0045202. It would be resolved by electrophysiology or synaptic-marker colocalization in LPA3 models — currently absent from the literature, so the annotation has no experimental foundation.
2. **Exact wording of the mouse GO:0010467 qualifier/relation in MGI.** Checked via QuickGO with/from = Q9EQ31 (IMP source

P 20864640).

Matters for whether to recommend recast vs. removal. Resolved by reading the MGI annotation record and the paper's full Results.

3. **Full text of**

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The seed states it contains PC12 Elk1 assays and tissue Northern blots but no synaptic assay; the abstract corroborates (LPA3 negative for PC12 MAP kinase). Full-text figure/methods confirmation would make the "remove GO:0007268" lead airtight.

4. **Human transfer of the axon-branching annotation.** The mouse record has axon IDA and `acts_upstream_of_or_within` neural-process terms; whether/how these are represented on human LPAR3 should be confirmed, as it affects what should replace the removed neural term.
5. **Remove-vs-recast for gene expression** is a curation-policy judgment call: whether a downstream transcriptional consequence merits any `acts_upstream_of_or_within` regulation of gene expression annotation, or none at all.

Discriminating Tests

- **Text/figure audit of**

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for any synapse/transmission assay (expected: none) → decides GO:0007268 removal.

- **Trace the GO:0045202 inference chain** in QuickGO (confirmed with/from=GO:0007268) → confirms the CC is dependent, not evidential.
 - **Read the MGI record for mouse Lpar3 GO:0010467** to confirm relationship type and whether GO already uses a regulation term → decides recast vs. remove.
 - **Synaptic electrophysiology in LPA3 loss/gain-of-function neurons** (mEPSC/mIPSC frequency & amplitude, paired-pulse ratio) — a positive result would be the only thing that could justify GO:0007268/GO:0045202.
 - **Subcellular localization of LPA3 in mature neurons** (synaptic markers synaptophysin/PSD-95 vs. axonal markers) — would test the synapse CC independently of the BP inference.
 - **Direct-vs-upstream test for gene expression:** nascent-RNA / RNA-Pol II occupancy time-course after LPA3-specific agonist (2(S)-OMPT) with TF-pathway inhibitors — would formally establish "upstream of," not "participates in," gene expression.
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Curation Leads (require curator verification)

1. Remove GO:0007268 (chemical synaptic transmission), involved_in, TAS

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— reference contains no synaptic assay; verify snippet: *"In PC12 cells, EDG4 but not EDG2 or EDG7 mediated the activation of MAP kinase by LPA."*

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2. Remove GO:0045202 (synapse), IEA GO_REF:0000108 — auto-inferred from GO:0007268 (with/from = GO:0007268); not independent.

3. Remove/recast GO:0010467 (gene expression), involved_in, IEA GO_REF:0000107 — ortholog transfer of mouse IMP

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verify snippet: *"These results demonstrate that pathways downstream of LPA3 are involved in the dynamic remodeling of ECM"*

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If retained at all, use **regulation of gene expression (GO:0010468)** with **acts_upstream_of_or_within**, flagged non-core.

4. Add/strengthen neural lead: neuron projection / axon branch formation from

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verify snippet: *"The LPA(3) agonist 2(S)-OMPT or LPA also induced axonal branch formation in hippocampal neurons, which was blocked by G(q) and Rnd2 pathway inhibition or LPA(3) knockdown."*

5. Suggested question for curators: Are the TAS synaptic-transmission and ortholog-transferred gene-expression annotations present across the LPAR paralog family (blanket-annotation check)? (Comparator result: no — LPAR3-only.)

Caveat: All recommendations are leads. The seed rightly warns against overriding experimental curators from abstract-only evidence; the two "remove" leads for GO:0007268 and GO:0010467 rest on (a) a TAS whose reference demonstrably lacks the assay and (b) an IEA whose upstream IMP is a downstream-transcription phenotype — both annotation-provenance problems rather than contradictions of experimental data, which strengthens the case for correction.

Bottom line

All three LPAR3 (Q9UBY5) annotations under review are over-annotations. *Chemical synaptic transmission* (GO:0007268) is a TAS to a cloning paper that performed no synaptic assay, and *synapse* (GO:0045202) is only an automatic inference from it — both should be removed. *Involved_in gene expression* (GO:0010467) is an ortholog transfer of a mouse mutant phenotype that is downstream of LPA3 signaling, and should be removed or recast as upstream regulation. The genuine, distinct neural role for LPA3 is axon localization and axon branch formation

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not synaptic transmission.

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